§5 简谐振动的合成

一、同一直线上同频率的两个简谐振动的合成

$$x_1 = A_1\cos(\omega t + \varphi_1) \qquad x_2 = A_2\cos(\omega t + \varphi_2)$$

合位移:

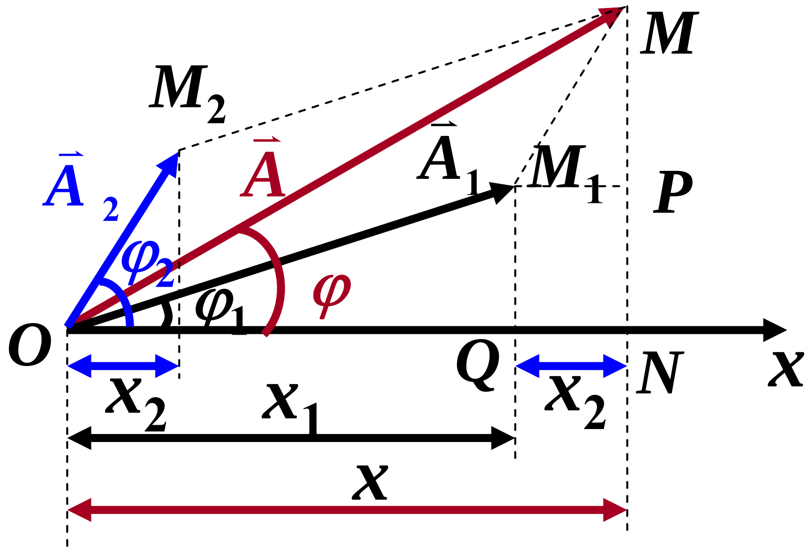
$$x = x_1 + x_2 = A_1\cos(\omega t + \varphi_1) + A_2\cos(\omega t + \varphi_2)$$

$$x = A\cos(\omega t + \varphi)$$

$$\begin{cases} A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_2 - \varphi_1)} \\[2ex] \tan\varphi = \dfrac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2} \end{cases}$$

$$x_1 = A_1 \cos(\omega t + \varphi_1)$$

$$x_2 = A_2 \cos(\omega t + \varphi_2)$$



旋转矢量法 $x = x_1 + x_2$

$\triangle OM_1M$ 中

$$\angle OM_1M = \pi - (\varphi_2 - \varphi_1)$$

$$A^2 = A_1^2 + A_2^2 - 2A_1A_2\cos[\pi - (\varphi_2 - \varphi_1)]$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_2 - \varphi_1)}$$

$$x = A\cos(\omega t + \varphi)$$

$$\tan\varphi = \frac{MN}{ON} = \frac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2}$$

$$x_1 = A_1\cos(\omega t + \varphi_1) \qquad x_2 = A_2\cos(\omega t + \varphi_2)$$

$$x = x_1 + x_2 = A\cos(\omega t + \varphi)$$

$$\begin{cases} A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_2 - \varphi_1)} \\ \tan\varphi = \dfrac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2} \end{cases}$$

推论：

同一直线上n个简谐振动所合成的运动仍为频率相同的简谐运动

讨论： $A=\sqrt{A_1^2+A_2^2+2A_1A_2\cos(\varphi_2-\varphi_1)}$

- $\left|A_1-A_2\right|\leq A\leq A_1+A_2$

- 若 $\varphi_2-\varphi_1=2k\pi \quad k=0,\pm1,\pm2\ldots$

  则 $\cos(\varphi_2-\varphi_1)=1$

  $A=\sqrt{A_1^2+A_2^2+2A_1A_2}=A_1+A_2=A_{max}$

  若 $A_1=A_2 \qquad A_{max}=2A_1$

讨论： $A=\sqrt{A_1^2+A_2^2+2A_1A_2\cos(\varphi_2-\varphi_1)}$

❖ 若 $\varphi_2-\varphi_1=(2k+1)\pi$　　$k=0,\pm1,\pm2\ldots$

则 $\cos(\varphi_2-\varphi_1)=-1$

$$A=\sqrt{A_1^2+A_2^2-2A_1A_2}=|A_1-A_2|=A_{\min}$$

若 $A_1=A_2$　　$A_{\min}=0$

振动互相抵消，质点处于静止状态。

相差对振动合成起着重要作用

例：某质点同时参与两个同方向、同频率的简谐振动

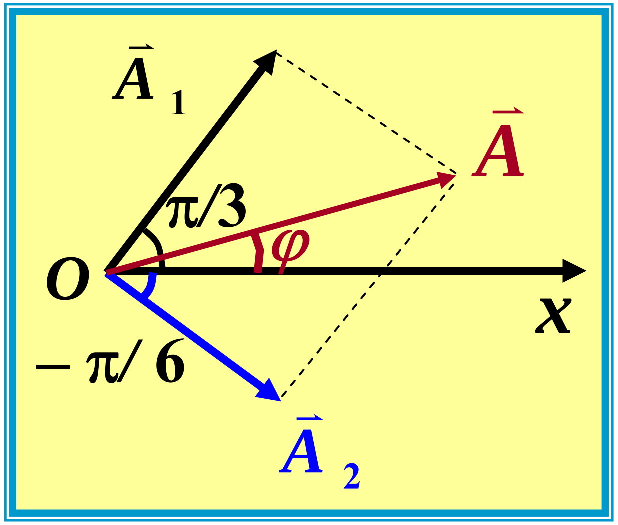
$$x_1 = 0.4\cos(3t+\frac{\pi}{3}),\quad x_2 = 0.3\cos(3t-\frac{\pi}{6})$$

求：(1)合振动的表达式

(2) 若 $x_3 = 0.5\cos(3t+\varphi_3)$

φ_3 为多少时，三者合振幅最大？

φ_3 为多少时，三者合振幅最小？



解：(1) $\varphi_1 - \varphi_2 = \pi/2$

$$A = \sqrt{A_1^2 + A_2^2} = 0.5(\mathrm{m}) \qquad \angle A_1OA = \alpha$$

$$\tan\alpha = \frac{0.3}{0.4} \qquad \alpha = 0.21\pi \qquad \varphi = \pi/3 - \alpha = 0.12\pi$$

$$A = 0.5(\mathrm{m}) \qquad \varphi = \pi/3 - \alpha = 0.12\pi$$

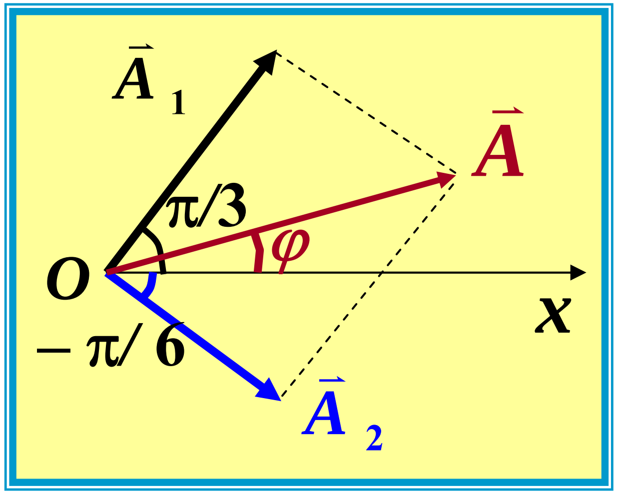
$$x = 0.5\cos(3t + 0.12\pi)$$

(2) $\varphi_3 = \varphi = 0.12\pi$,

$$A_{\max} = A_1 + A_2 = 0.5 + 0.5 = 1.0(\mathrm{m})$$

(3) $\varphi_3 = \varphi + \pi = 1.12\pi$,

$$A_{\min} = |A_1 - A_2| = 0.5 - 0.5 = 0$$



二、同一直线上 n 个同频率的简谐振动的合成

$$x_1 = a\cos\omega t$$

$$x_2 = a\cos(\omega t + \delta)$$

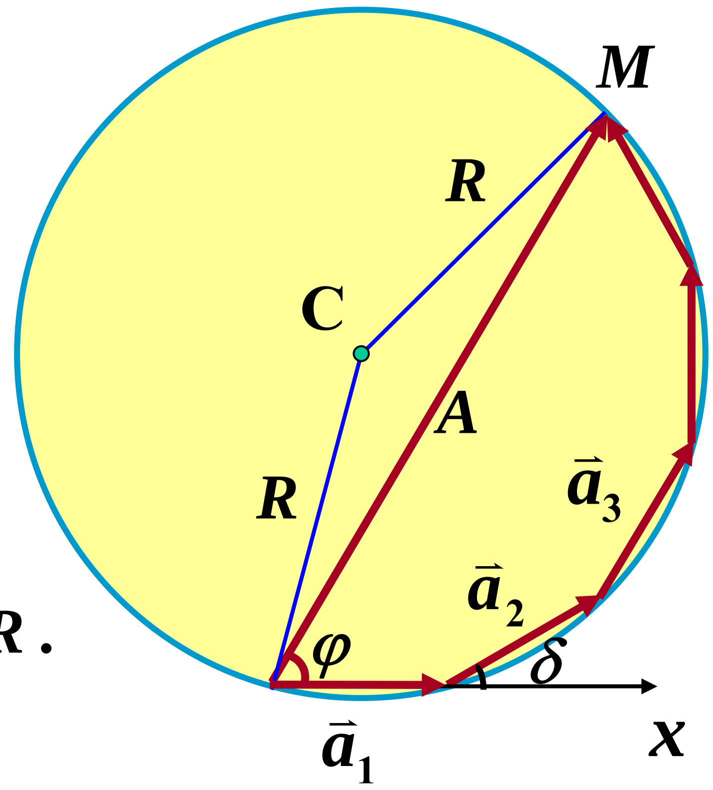
$$x_3 = a\cos(\omega t + 2\delta)$$

$$\cdots$$

$$x_n = a\cos[\omega t + (n-1)\delta]$$

C: 正多边形外接圆圆心，半径 R.

$$x = A\cos(\omega t + \varphi)$$



$\angle PCQ = \delta \qquad \angle OCN = \delta \qquad \angle MCO = n\,\delta$

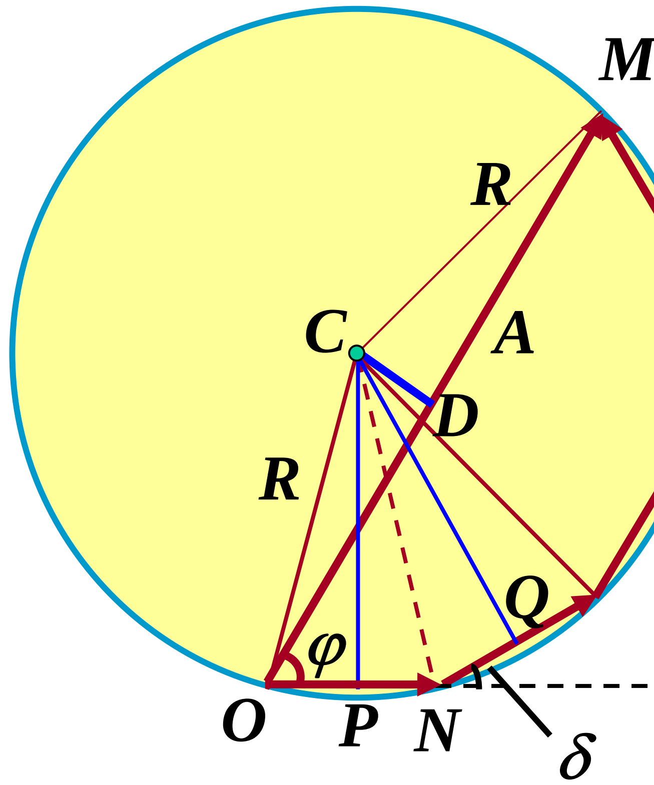
Δ MCO中 $A = 2\cdot R\sin\dfrac{n\delta}{2}$

Δ OCN中 $a = 2\cdot R\sin\dfrac{\delta}{2}$

$$A = a\frac{\sin(n\delta/2)}{\sin(\delta/2)}$$

$\angle COM = \dfrac{1}{2}(\pi - n\delta) \qquad \angle CON = \dfrac{1}{2}(\pi - \delta)$

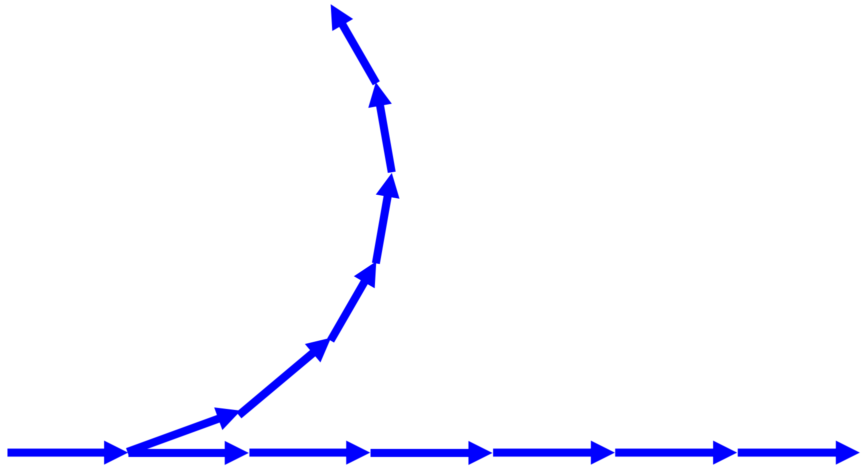
$\varphi = \angle CON - \angle COM = \dfrac{1}{2}(n-1)\delta$



$$A = a\frac{\sin(n\delta/2)}{\sin(\delta/2)} \qquad \varphi = \frac{1}{2}(n-1)\delta$$

$$x = A\cos(\omega t+\varphi) = a\frac{\sin(n\delta/2)}{\sin(\delta/2)}\cos\left(\omega t+\frac{n-1}{2}\delta\right)$$

讨论：$A = a\frac{\sin(n\delta/2)}{\sin(\delta/2)}$

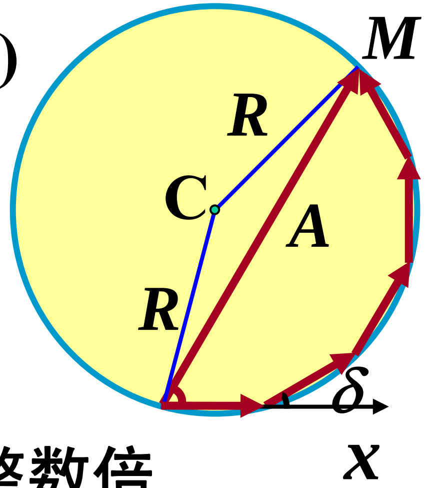


☺ 若 $\delta = 2k\pi$，$k = 0, \pm 1, \pm 2 \ldots$

$A = na$

最大合振幅

$$x = A\cos(\omega t+\varphi) = a\frac{\sin(n\delta/2)}{\sin(\delta/2)}\cos(\omega t+\frac{n-1}{2}\delta)$$



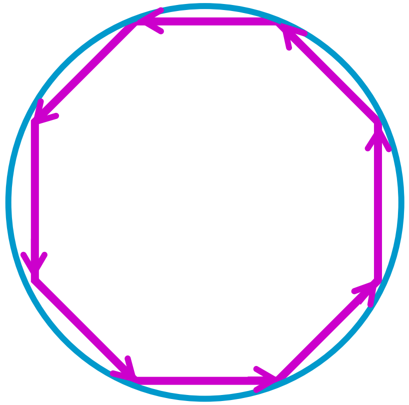
讨论： $A = a\dfrac{\sin(n\delta/2)}{\sin(\delta/2)}$

☺ 若 $\delta = \dfrac{2k'\pi}{n}$ k'为非零整数，但 k' 不为 n 的整数倍

$k' = \pm 1, \pm 2 \ldots, \pm(n\text{-}1), \pm(n+1), \ldots, \pm(2n\text{-}1), \pm(2n+1),$

…

$n\delta = 2k'\pi$ $A = 0$ 合振幅为零



结论

$$x_1 = a\cos\omega t \qquad x_2 = a\cos(\omega t+\delta) \qquad x_3 = a\cos(\omega t + 2\delta)$$

$$\cdots \qquad x_n = a\cos[\omega t + (n-1)\delta]$$

$$x = A\cos(\omega t+\varphi) = a\frac{\sin(n\delta/2)}{\sin(\delta/2)}\cos(\omega t + \frac{n-1}{2}\delta)$$

☺ 若 $\delta = 2k\pi$，

$k = 0, \pm 1, \pm 2 \ldots$

$A = na$

最大合振幅

☺ 若 $\delta = \dfrac{2k'\pi}{n}$

$k' = \pm 1, \pm 2 \ldots, \pm(n-1), \pm(n+1)$,

$\ldots$,

$\pm(2n-1), \pm(2n+1), \ldots$

$A = 0$ 合振幅为零

三、同一直线上不同频率的简谐振动的合成

$\vec{A}_1 \quad x_1 = A\cos(\omega_1 t + \varphi_1), \quad \vec{A}_2 \quad x_2 = A\cos(\omega_2 t + \varphi_2)$

$$\Delta\varphi = (\omega_2 t + \varphi_2) - (\omega_1 t + \varphi_1) = (\omega_2 - \omega_1)t + (\varphi_2 - \varphi_1)$$

合振动 $x = x_1 + x_2 = A\cos(\omega_1 t + \varphi_1) + A\cos(\omega_2 t + \varphi_2)$

若 $\varphi_1 = \varphi_2 = 0$, $\quad x = 2A\cos(\dfrac{\omega_2 - \omega_1}{2})t \cdot \cos(\dfrac{\omega_2 + \omega_1}{2})t$

合振动不是简谐振动

$$x = 2A\cos(\frac{\omega_2 - \omega_1}{2})t \cdot \cos(\frac{\omega_2 + \omega_1}{2})t$$

特例：若ω_2 与 ω_1都较大，但是数值相近

设$\omega_2 > \omega_1 (\nu_2 > \nu_1)$，$\omega_2 - \omega_1 << \omega_2 + \omega_1$

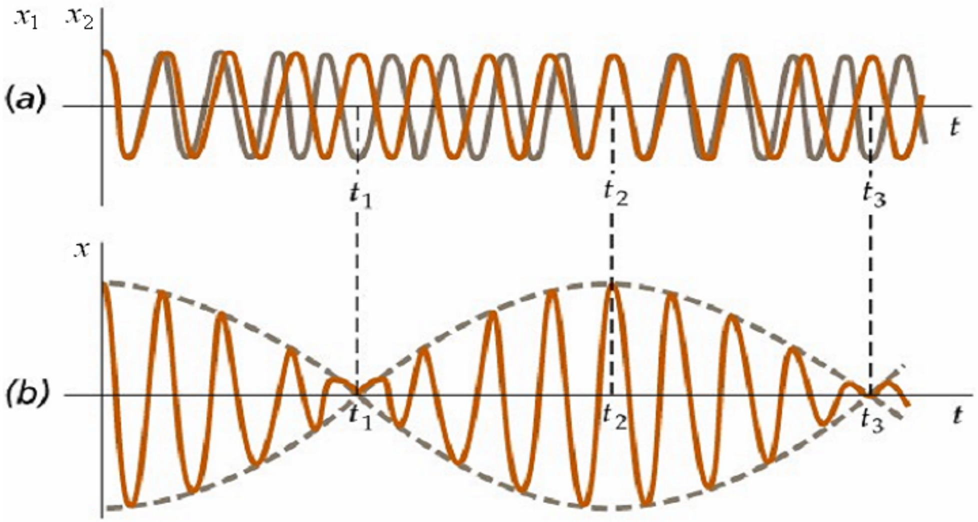
$$A(t) = 2A\cos(\frac{\omega_2 - \omega_1}{2})t \quad \text{随 } t \text{ 缓变}$$

$$x = A(t)\cos\overline{\omega}t$$

$$\cos\overline{\omega}t = \cos(\frac{\omega_2 + \omega_1}{2}t) \quad \text{随 } t \text{ 快变}$$

合振动可看作振幅缓变的简谐振动 $\left|2A\cos(\frac{\omega_2 - \omega_1}{2})t\right|$

平均圆频率 $\omega_{平均} = \dfrac{\omega_2 + \omega_1}{2}$　　调制圆频率 $\omega_{调制} = \dfrac{\omega_2 - \omega_1}{2}$



$$x = A(t)\cos\overline{\omega}t$$

$$A(t) = 2A\cos(\frac{\omega_2 - \omega_1}{2})t$$

$$\cos\overline{\omega}t = \cos(\frac{\omega_2 + \omega_1}{2}t)$$

振动方向相同、频率之和远远大于频率之差的两个简谐振动合成时，合振动振幅呈周期性变化的现象称为拍。

合振幅变化一个周期为一拍。

单位时间内拍出现的次数为拍频。

特例：若ω_2 与 ω_1 相近(都较大)。设 $\omega_2 > \omega_1$， $\nu_2 > \nu_1$

考虑相量图 (旋转矢量图)

$\vec{A}_1$ 单位时间内转ν_1圈

$\vec{A}_2$ 单位时间内转ν_2圈

单位时间内， 2比1多转$(\nu_2 - \nu_1)$圈

$\vec{A}_1$ $\vec{A}_2$

同向重合 $(\nu_2 - \nu_1)$次 振动加强$(\nu_2 - \nu_1)$次

相背$(\nu_2 - \nu_1)$次 振动减弱$(\nu_2 - \nu_1)$次

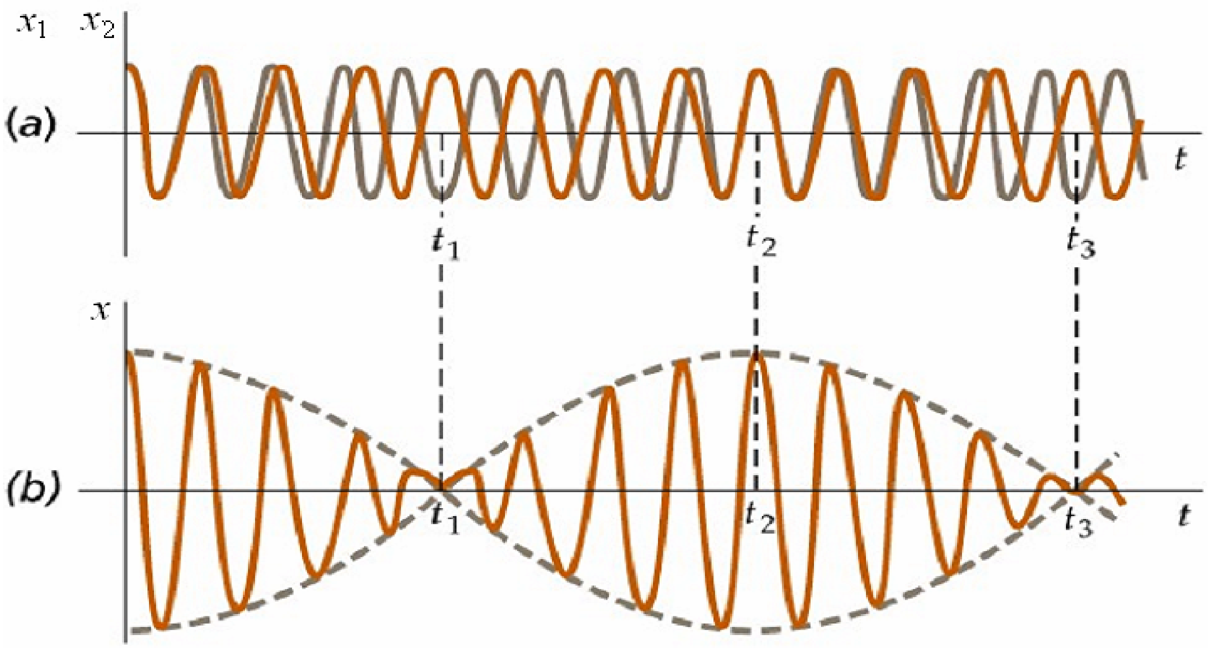
形成拍

拍频ν:单位时间内振动加强或减弱的次数

$$\nu = (\nu_2 - \nu_1)$$

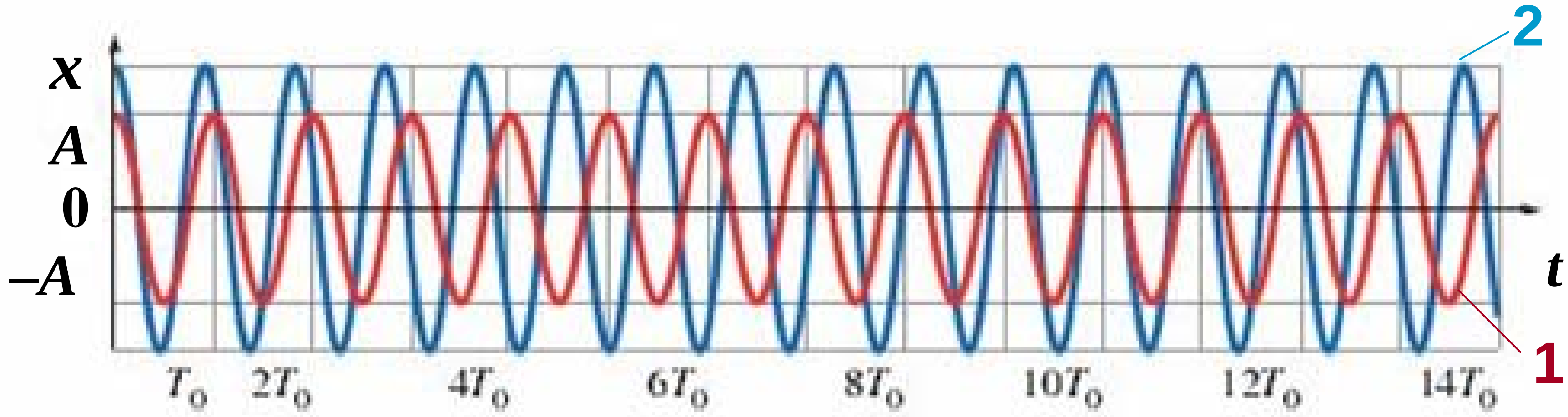
结论：同一直线上不同频率的简谐振动的合成

在ω_2 与 ω_1都较大，且数值相近的条件下，合振动可视为振幅缓慢变化的简谐振动，合振动的振幅周期性变化，出现拍的现象。



拍频：$\nu = |\nu_2 - \nu_1|$

拍的计算



$\nu_1 = 1/T_0$ $\nu_2 = 1.1\,\nu_1$ $\nu_2 = 1.1/T_0$

拍频： $\nu = 0.1/T_0 = 1/(10T_0)$

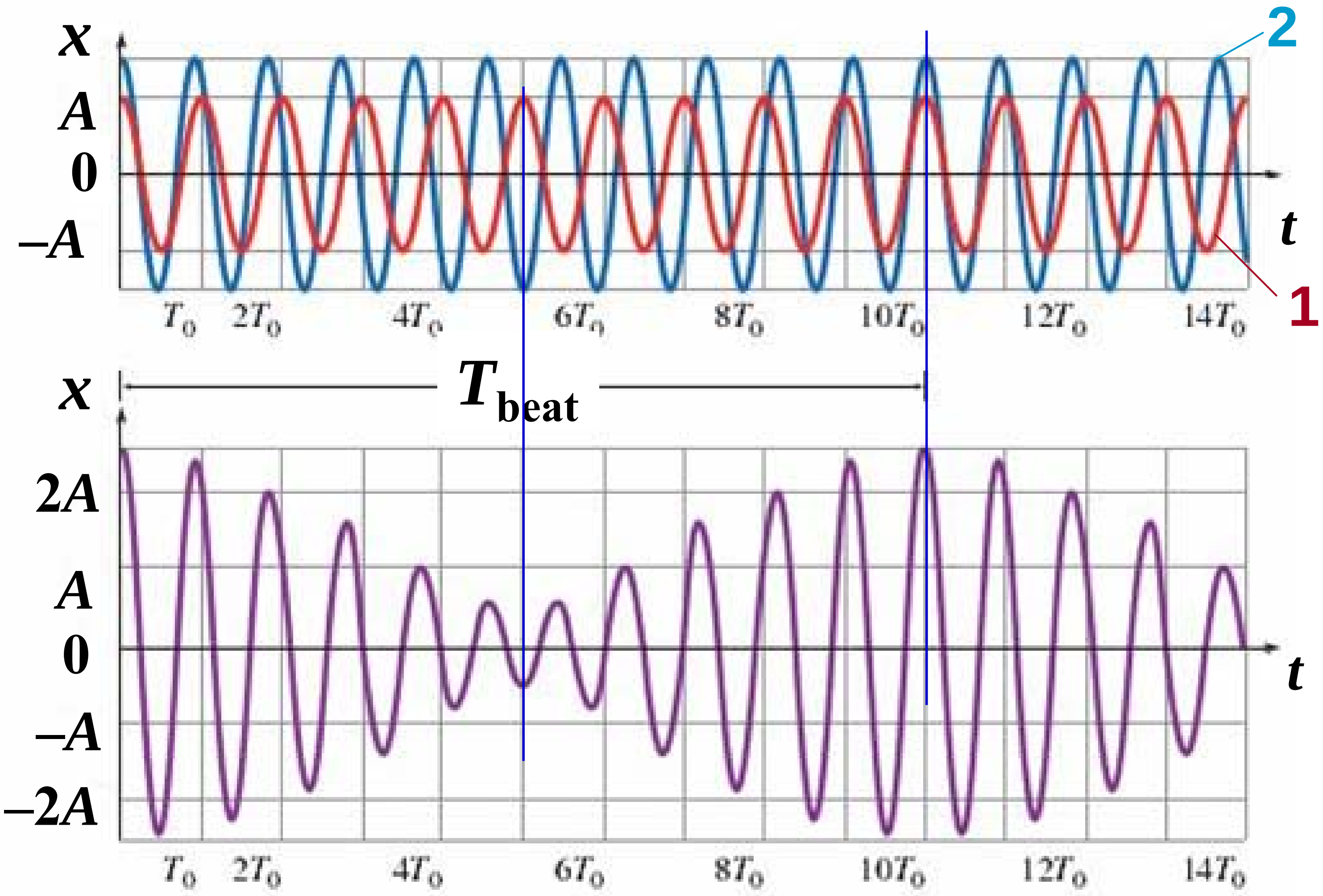
拍的计算

$\nu_1 = 1/T_0$

$\nu_2 = 1.1\,\nu_1$

$\nu_2 = 1.1/T_0$

$\nu = 1/(10T_0)$



例如：钢琴调音师敲击音叉（$\nu = 523.3\text{Hz}$），同时敲击一个钢琴琴键。两者频率近似相等，他听到了每秒有3.0个拍。逐渐张紧琴弦，他听到拍频慢慢降为每秒2.0个拍。已知使琴弦张紧会提高弦的频率，求琴弦张紧前频率ν_0是多少？

解：拍频等于两个频率的差

$$\Delta\nu = 3.0\text{Hz}$$

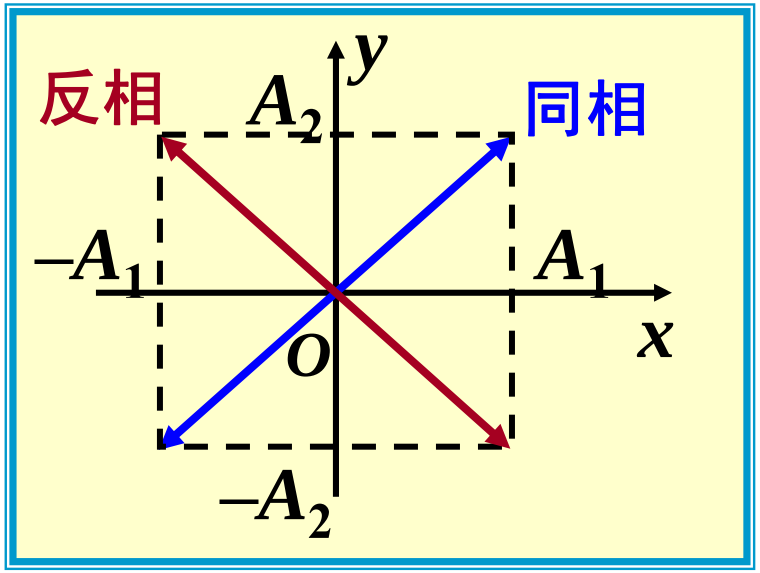
$$\nu_0 = 523.3\text{Hz} - 3.0\text{Hz} = 520.3\text{Hz}$$

四、相互垂直的简谐振动的合成

1. 同周期

$$x = A_1 \cos(\omega t + \varphi_1)$$

$$y = A_2 \cos(\omega t + \varphi_2)$$



（1）$\varphi_2 - \varphi_1 = 0$ 同相 $\varphi_2 = \varphi_1 = \varphi$

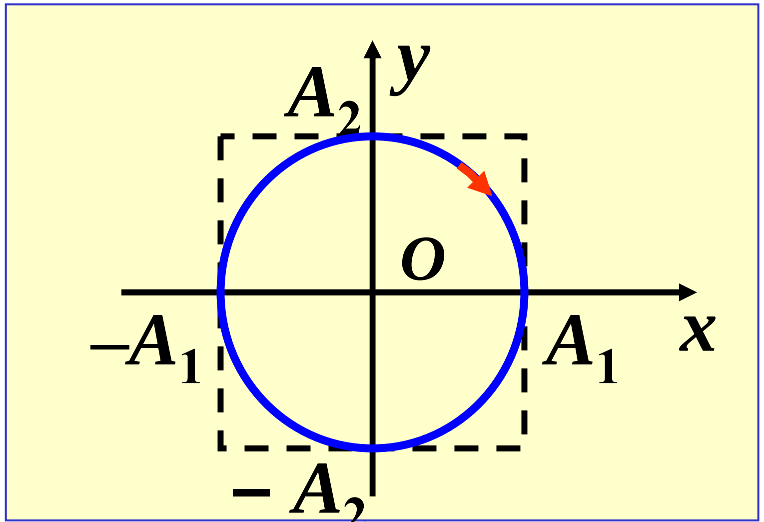
$$\frac{x}{A_1} = \frac{y}{A_2} \qquad y = \frac{A_2}{A_1} x$$

质点离开原点的位移s为：$s = \sqrt{A_1^2 + A_2^2} \cos(\omega t + \varphi)$

合振动为简谐振动

（2）$\varphi_2 - \varphi_1 = \pi$ 反相 $y = -\dfrac{A_2}{A_1} x$

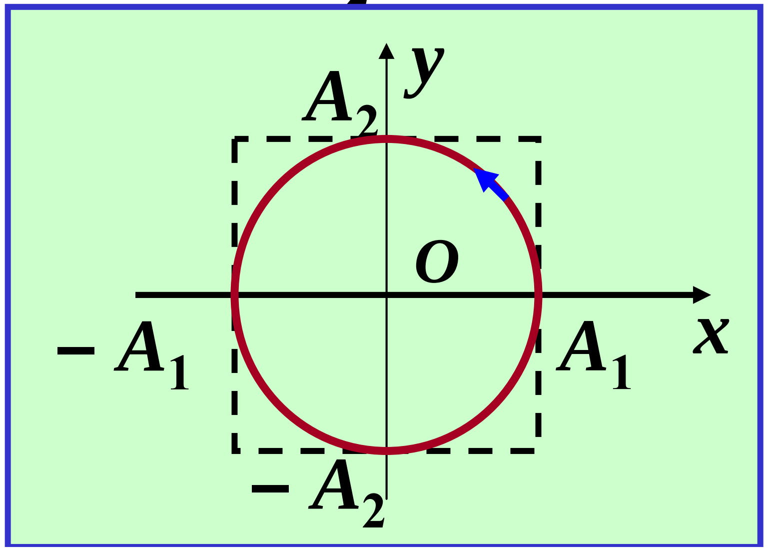
$x = A_1\cos(\omega t + \varphi_1)$ $\quad y = A_2\cos(\omega t + \varphi_2)$



（3）$\varphi_2 - \varphi_1 = \dfrac{\pi}{2}$ $\quad y$ 超前 x

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1$$

轨迹：以坐标轴为轴的正椭圆
顺时针，右旋



（4）$\varphi_2 - \varphi_1 = -\dfrac{\pi}{2}$ $\quad x$ 超前 y

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1$$

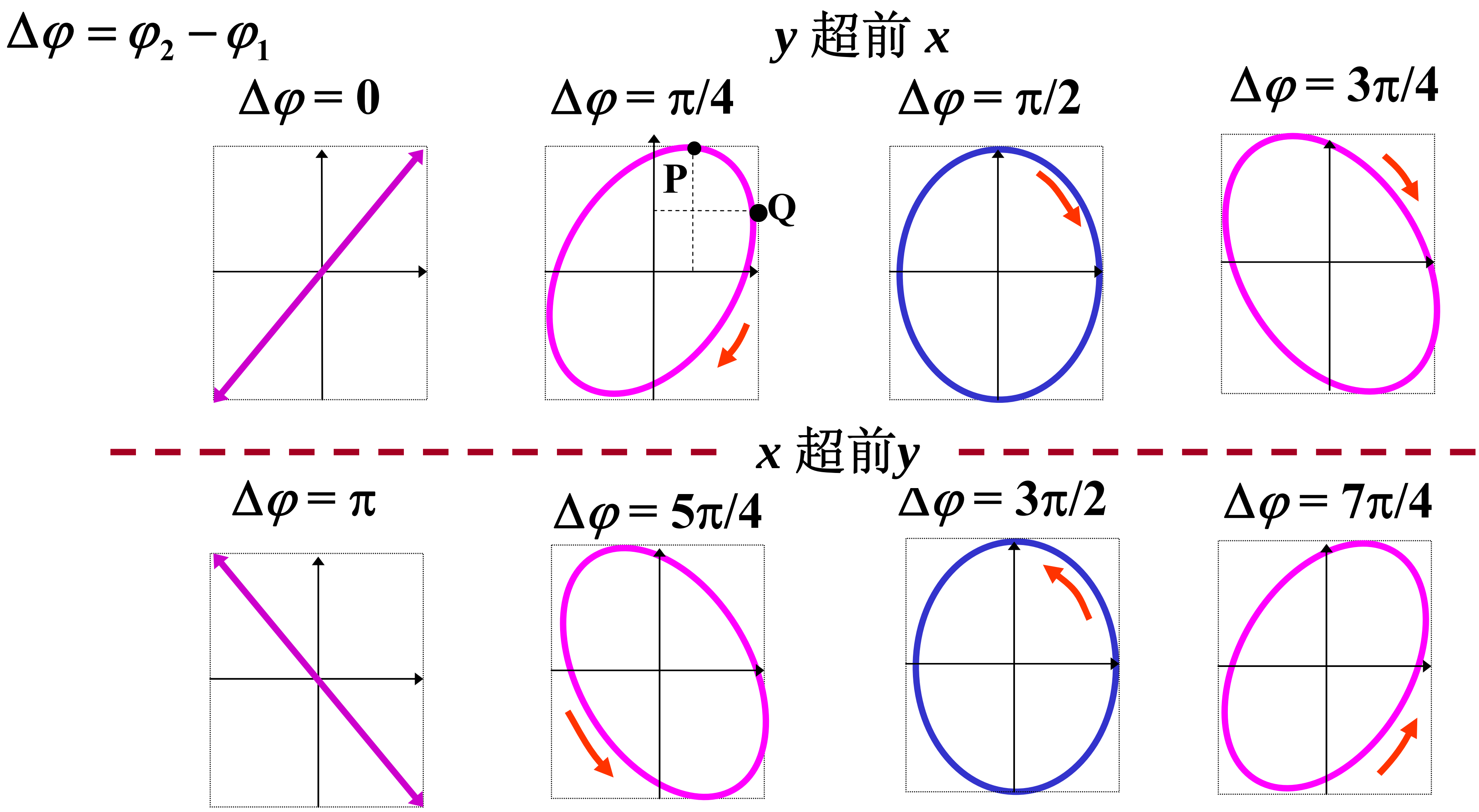
逆时针，左旋

$$x = A_1 \cos(\omega t + \varphi_1) \qquad y = A_2 \cos(\omega t + \varphi_2)$$

（5） 一般 $\Delta\varphi = \varphi_2 - \varphi_1$

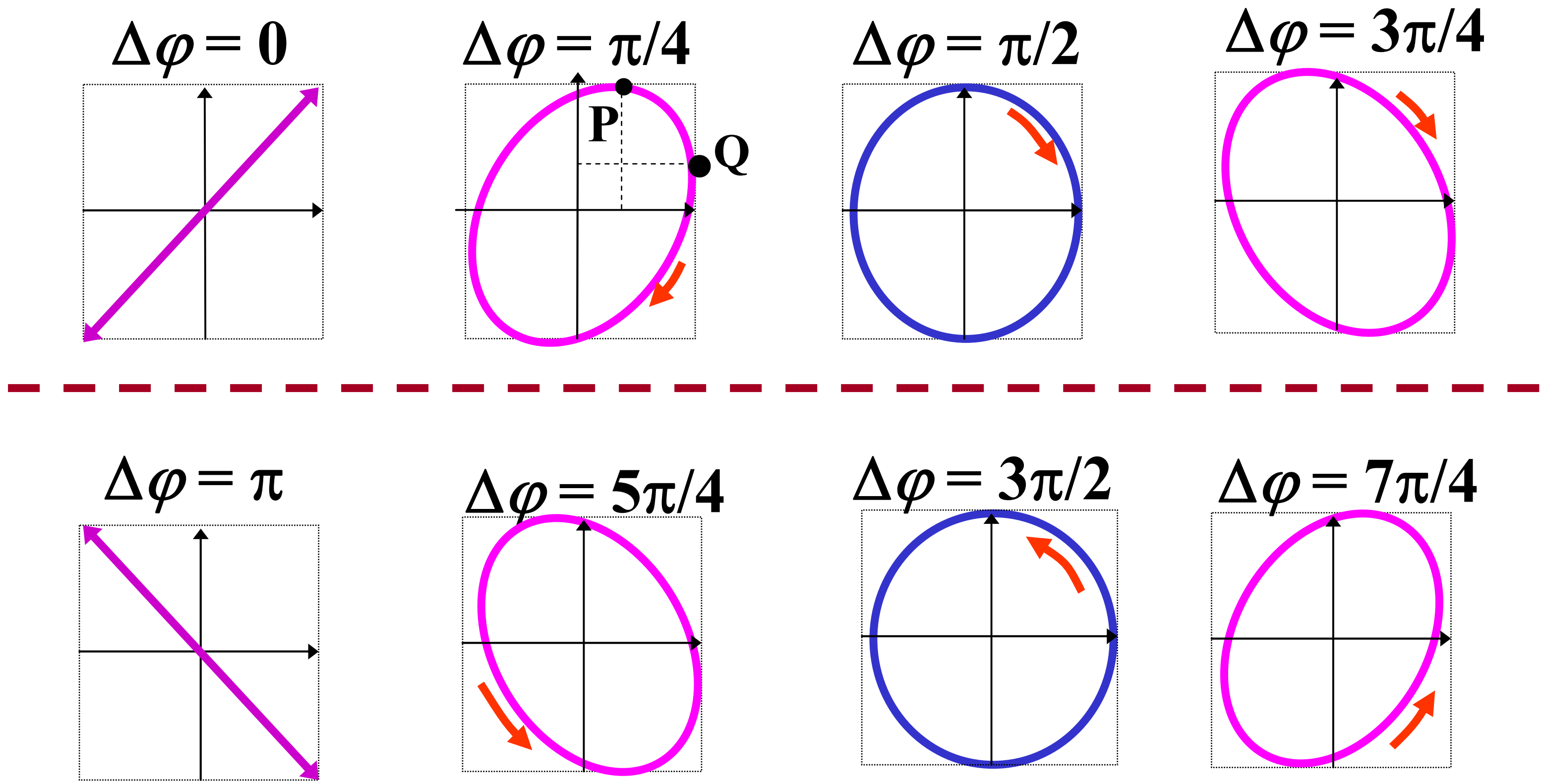
$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - \frac{2xy}{A_1 A_2}\cos(\varphi_2 - \varphi_1) = \sin^2(\varphi_2 - \varphi_1)$$

轨迹为椭圆


$\Delta\varphi = \varphi_2 - \varphi_1$
y 超前 x
$\Delta\varphi = 0$
$\Delta\varphi = \pi/4$
P
Q
$\Delta\varphi = \pi/2$
$\Delta\varphi = 3\pi/4$
x 超前 y
$\Delta\varphi = \pi$
$\Delta\varphi = 5\pi/4$
$\Delta\varphi = 3\pi/2$
$\Delta\varphi = 7\pi/4$

2. 相互垂直、不同周期简谐振动的合成

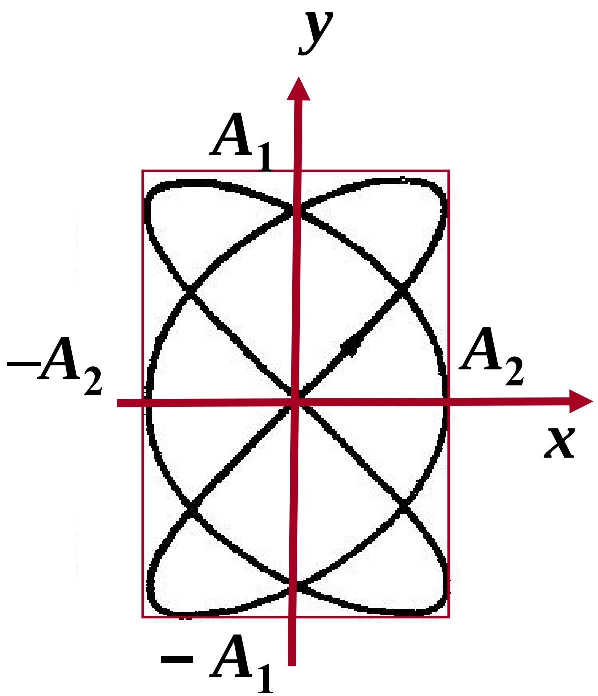
（1） 两振动频率相差很小



(2) 两振动频率相差较大，但有简单的整数比

$$\frac{\omega_x}{\omega_y}=\frac{m}{n},\quad m,n \text{ 正整数}$$

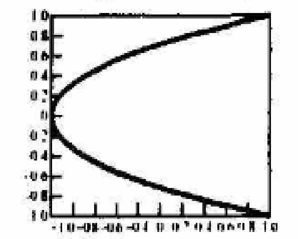
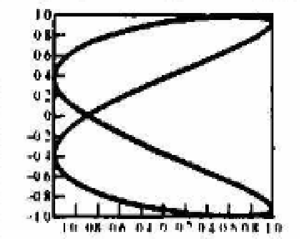
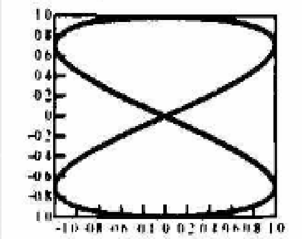
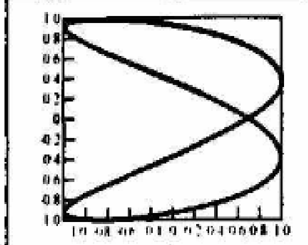
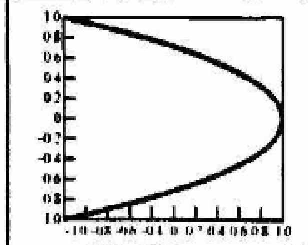
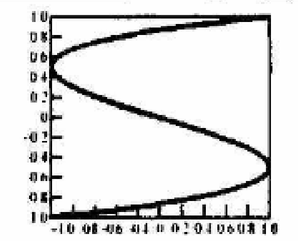
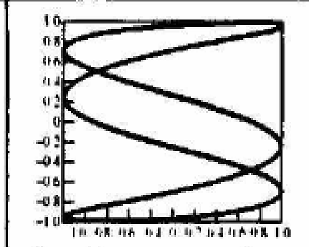
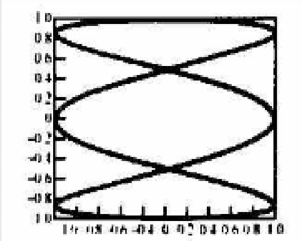
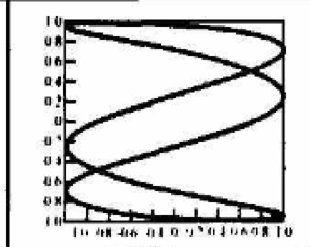
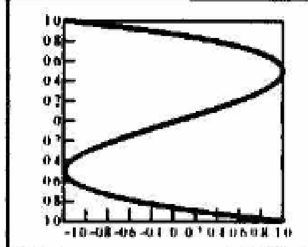
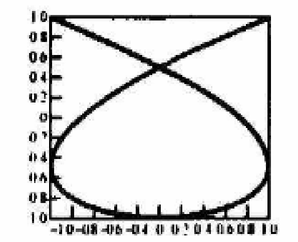
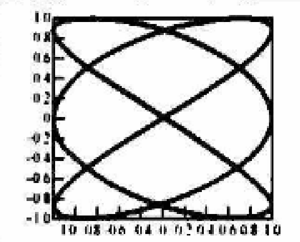
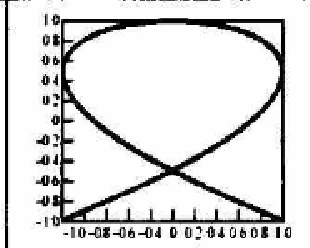
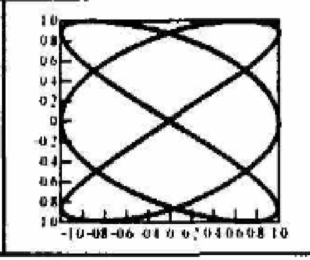
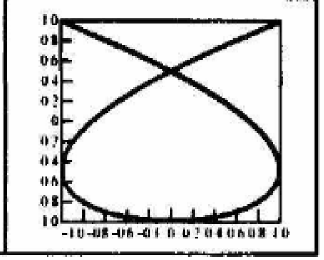
合成轨迹为稳定的曲线—李萨如图



如图：$$\frac{\omega_x}{\omega_y}=\frac{3}{2}$$

应用：测定未知频率

简单频率比与初相差的李萨如图形

ω_x/ω_y \ $\varphi_x-\varphi_y$	0	π/4	π/2	3π/4	π
2:1					
3:1					
3:2					

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